Implementation of Unbounded CFFs

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Introduction

A digital signature is a cryptographic proof of integrity and authenticity.

Problem: If n documents were digitally signed, how can we verify all of them efficiently?

- ▷ **Naive solution:** verify each one individually;
- ▷ **Good solution:** aggregate all signatures into one and perform one verification;
- ▷ **Better solution:** fault-tolerant aggregation of signatures using d-CFFs;

Definition 1. *A $t \times n$ binary matrix M is a d-CFF(t, n) if the union of any d columns does not cover any other column.*

$$M = \begin{array}{c} a_1 \\ a_2 \\ a_3 \\ a_4 \end{array} \begin{array}{|cccccc|} 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{array}$$

1-CFF(4,6)

- ▷ Columns: individual signatures;
- ▷ Rows: aggregation of corresponding signatures;
- ▷ $n = 6$ signatures/documents;
- ▷ $t = 4$ aggregations and verifications;
- ▷ Fault-tolerance $d = 1$;

Problem: How can we add more documents without restructuring the entire schema?

Solution: Infinite sequence of CFFs, each d-CFF M^l is embedded into d'-CFF M^{l+1}.

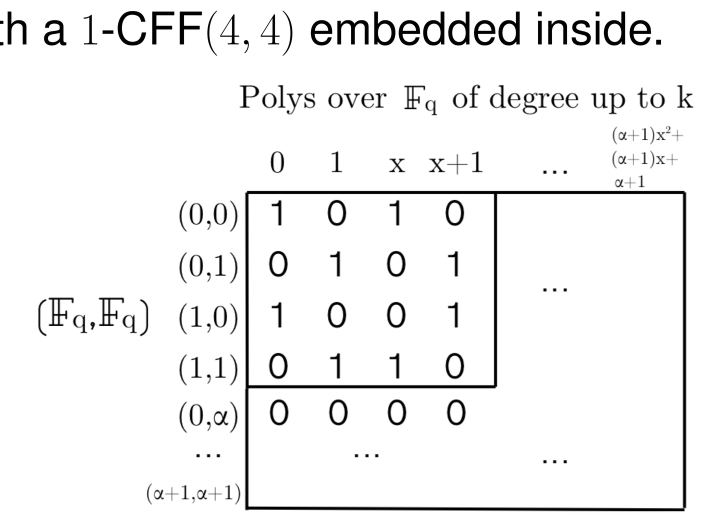
- ▷ We call such a sequence of CFFs an *embedding family* of CFFs.
- ▷ Original aggregations are preserved as the CFF grows.
- ▷ Construction based on polynomials over finite fields.

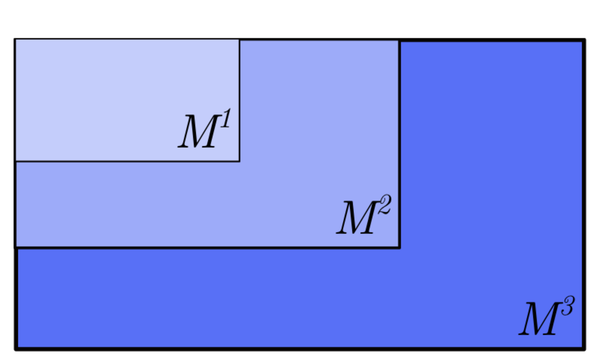
Theorem 1. *[1, 2] Let q be a prime power, $k \geq 1$. Then, there exists a d-CFF(q^2, q^{k+1}), with $d \leq \frac{q-1}{k}$.*

Notation: $M_{q,k,d}$ denotes a d-CFF(q^2, q^{k+1}).

Theorem 2. *[2] Let q be a prime power, $q \geq d_0 k_0 + 1$. Let $k_i \leq k_{i+1}$, $d_i \leq d_{i+1}$, $q^{2^i} \geq d_i k_i + 1$, for all $i \geq 0$. Then, the sequence $\{M_{q^{2^i}, k_i, d_i}\}_{i \geq 0}$ forms an embedding family of CFFs.*

- ▷ Let $q = 2$, $\mathbb{F}_2 = \{0, 1\}$ with $k_0 = 1$, $d_0 = 1$, we build a 1-CFF$(4, 4)$ $M_{2,1,1}$.
- ▷ Extending to $q^2 = 4$, $\mathbb{F}_4 = \{0, 1, \alpha, \alpha + 1\}$ with $k_1 = 2$, $d_1 = 1$, we obtain a 1-CFF$(16, 64)$ $M_{4,2,1}$ with a 1-CFF$(4, 4)$ embedded inside.





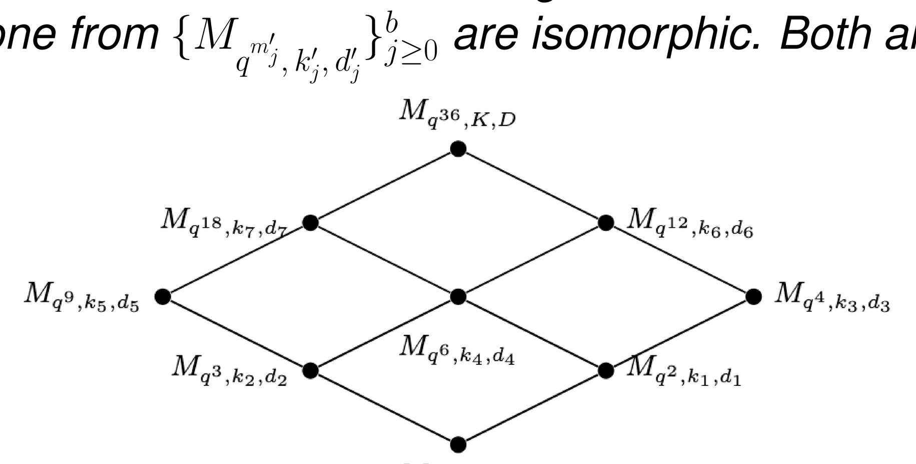
1. Generalization

We generalize the construction from [2] by considering that a finite field $\mathbb{F}_m$ can be extended to $\mathbb{F}_n$ if and only if $m \mid n$ [4]. Using this property, the construction of embedding CFFs can be generalized as follows.

Theorem 3. *Let q be a prime power, $(m_i)_{i \geq 0}$ a sequence of integers with $m_i \geq 1$ and $m_i \mid m_{i+1}$ for all $i \geq 0$. Let $(k_i)_{i \geq 0}$ and $(d_i)_{i \geq 0}$ be nondecreasing sequences of positive integers, and assume $q^{m_i} \geq d_i k_i + 1$ for all $i \geq 0$. Then, the sequence $\{M_{q^{m_i}, k_i, d_i}\}_{i \geq 0}$ forms an embedding family of CFFs.*

This allows the embedding family of CFFs to grow through different paths. The next theorem guarantees that any two such paths reaching the same final parameters yield isomorphic CFFs.

Theorem 4. *Let q be a prime power. Let $(m_0, m_1, \ldots, m_a)$ and $(m'_0, m'_1, \ldots, m'_b)$ be two different sequences of positive integers, such that $m_i \mid m_{i+1}$ for all $i \in \{0, \ldots, a-1\}$, $m'_j \mid m'_{j+1}$ for all $j \in \{0, \ldots, b-1\}$, and $m_a = m'_b = M$. Let $(k_i)_{i=0}^a$, $(d_i)_{i=0}^a$, $(k'_j)_{j=0}^b$, and $(d'_j)_{j=0}^b$ be nondecreasing sequences, such that $k_a = k'_b = K$ and $d_a = d'_b = D$. Assume both conditions $q^{m_i} \geq d_i k_i + 1$ for all $i \in \{0, \ldots, a\}$, $q^{m'_j} \geq d'_j k'_j + 1$ for all $j \in \{0, \ldots, b\}$. Then, the final CFF resulting from the construction via sequence $\{M_{q^{m_i}, k_i, d_i}\}_{i \geq 0}^a$ and the one from $\{M_{q^{m'_j}, k'_j, d'_j}\}_{j \geq 0}^b$ are isomorphic. Both are matrix representations of $M_{q^M, K, D}$.*



- ▷ Following the construction from [2] we would only obtain a single sequence: $M_{q^1,k_0,d_0} \to M_{q^2,k_1,d_1} \to M_{q^4,k_3,d_3}$

- ▷ More flexibility in the applications.
- ▷ CFF can grow through different paths depending on how many items will be added.

2. Practical Implementation

Challenge: How to efficiently perform the qq^{k+1} polynomial evaluations to build the CFF?

Our approach: Two parallelizable phases: pre-compute evaluations, then build the CFF.

- ▷ **Inverted Index:** nested hash map $(x, P_j(x)) \mapsto \{j\}$, for $x \in \mathbb{F}_q$;
- ▷ **Build CFF Matrix:** fills M using the index, setting $M_{(x,y),j} = 1$ for each retrieved j;

```
procedure BuildInvertedIndex(𝒫, 𝔽_q)
    InvIndex ← new Map()
    for all x ∈ 𝔽_q do (parallel)
        InnerMap ← new Map()
        for all P_j ∈ 𝒫 do
            y ← P_j(x)
            if y is not a key in InnerMap then
                InnerMap.insert(y, new List())
            IndexList ← InnerMap.get(y)
            IndexList.append(j)
        critical: InvIndex.insert(x, InnerMap)
    return InvIndex
```

```
procedure BuildCFFMatrix(InvIndex, 𝒯, t, n)
    M ← new Matrix(t, n, fill_value = 0)
    for all i ∈ {1, ..., t} do (parallel)
        (x, y) ← 𝒯[i]
        InnerMap ← InvIndex.get(x)
        if InnerMap is not null then
            IndexList ← InnerMap.get(y)
            if IndexList is not null then
                for all j ∈ IndexList do
                    M[i][j] ← 1
    return M
```

Embedding:

```
procedure EmbeddingCFF(𝒫_old, 𝒫_new, 𝒯_old, 𝒯_new, C, 𝔽_q')
    Index_old ← BuildInvertedIndex(𝒫_old, 𝔽_q')
    Index_new ← BuildInvertedIndex(𝒫_new, 𝔽_q')
    M_Y ← BuildCFFMatrix(Index_new, 𝒯_old, |𝒯_old|, |𝒫_new|)
    M_Z ← BuildCFFMatrix(Index_old, 𝒯_new, |𝒯_new|, |𝒫_old|)
    M_W ← BuildCFFMatrix(Index_new, 𝒯_new, |𝒯_new|, |𝒫_new|)
    M ← [ C    M_Y ]
        [ M_Z  M_W ]
    return M
```

3. Experiments

- ▷ **Inv-CFF(s):** total time to build the inverted index and fill the CFF matrix.
- ▷ **CFF(s):** filling time with a precomputed index.
- ▷ Each row represents a CFF obtained by embedding the previous one.

q	k	d	t	n	Inv-CFF(s)	CFF(s)
2	1	1	4	4	0.000247	0.000059
4	1	3	16	16	0.000179	0.000065
4	2	1	16	64	0.000157	0.000023
16	2	7	256	4,096	0.007461	0.000980
16	3	5	256	65,536	0.526912	0.013431
16	4	3	256	1,048,576	110.564562	0.219198
3	1	2	9	9	0.000208	0.000050
3	2	1	9	27	0.000141	0.000022
9	2	4	81	729	0.000972	0.000138
9	3	2	81	6,561	0.014614	0.000481
9	4	2	81	59,049	0.682487	0.003866
9	5	1	81	531,441	50.117430	0.034134
5	1	4	25	25	0.000212	0.000041
5	2	2	25	125	0.000217	0.000023
25	2	12	625	15,625	0.035868	0.006657
25	3	8	625	390,625	10.113630	0.163782

q	k	d	t	n	Inv-CFF(s)	CFF(s)
2	1	1	4	4	0.000260	0.000066
4	1	1	8	16	0.000204	0.000070
16	1	1	32	256	0.000625	0.000076
256	1	1	512	65,536	3.074890	0.019248
3	1	2	9	9	0.000158	0.000047
9	1	2	27	81	0.000270	0.000076
81	1	2	243	6,561	0.063837	0.001273
3	2	1	9	27	0.000223	0.000061
27	2	1	81	19,683	0.047209	0.001513
5	2	2	25	125	0.000253	0.000054
25	2	2	125	15,625	0.035310	0.001686

- ▷ MacBook Air, Apple M2 (8-core CPU, 8 GB), $T = 8$ threads

4. Conclusion

- ▷ CFFs have several applications in cryptography, particularly in dynamic schemes that require new items to be added over time.

Contributions:

- ▷ Proposed a generalization for the construction of embedding families of CFFs.
- ▷ Developed an efficient implementation of this sequence of objects.

Future Work:

- ▷ Applying unbounded CFFs to practical d-CFF-based cryptographic protocols.
- ▷ Explore other constructions for embedding families with a smoother growth of n.

References

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[4] Gary L. Mullen and Daniel Panario, editors. *Handbook of Finite Fields*. Discrete Mathematics and Its Applications. CRC Press, Boca Raton, FL, 2013.

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